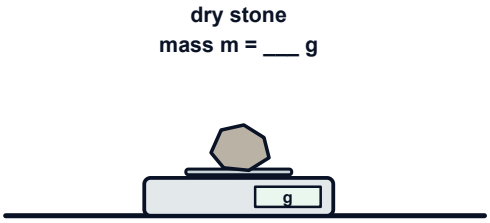
DO NOW, REQUIRED PRACTICAL

# Density of an irregular solid

**Particle model of matter**
Paper 1, Combined and Separate
6 steps

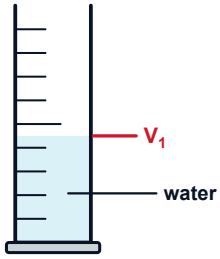
A student measures the density of a small stone.

## 1 Measure the mass of the dry stone with a balance.



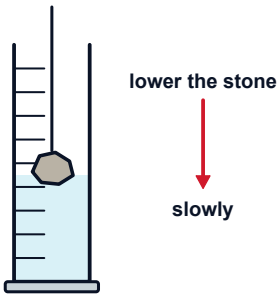

measure the mass

## 2 Part-fill a measuring cylinder with water and record the volume.

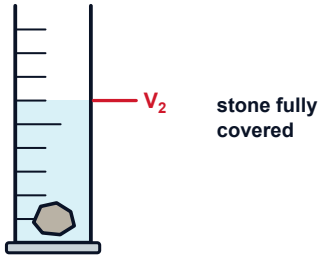



part-fill with water and record the volume $V_1$

## 3 Lower the stone into the water until it is fully covered.



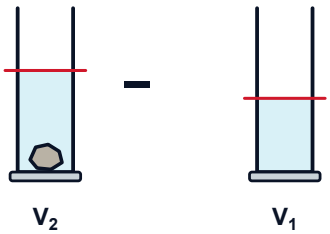

## 4 Record the new volume shown on the measuring cylinder.




record the new volume $V_2$

## 5 Subtract the two volumes to find the volume of the stone.

**volume of stone = $V_2 - V_1$**




## 6 Calculate the density: mass ÷ volume.

$$\rho = m \div V$$

density (g/cm³) = mass (g) ÷ volume (cm³)

m = mass of the stone, V = $V_2 - V_1$

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